

$$H \rightarrow WW + \text{charm}$$


Analysis status

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2022postEE · 26.7 fb⁻¹ · expected UL 1034

Sections

§	topic
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- | | |
|---|---|
| 1 | Analysis and selection |
| 2 | MVA-defined regions |
| 3 | c-tagging — 2D working points and scale factors |
| 4 | Negative-weight reweighting |
| 5 | W+jets jet-binned samples |
| 6 | Systematics and impacts |
| 7 | Status and outlook |
-

1 · Analysis and selection



Why $H+c$

The Higgs coupling to **charm** is the least constrained of the accessible Yukawas. Associated production $H + c$ probes it directly, without relying on the $H \rightarrow c\bar{c}$ decay.

production	$pp \rightarrow H + c (+X)$
decay	$H \rightarrow WW^* \rightarrow 2\ell 2\nu$
final state	opposite-sign $e\mu + \text{MET} + \geq 1$ c-tagged jet
dataset	2022postEE, 26.7 fb⁻¹ , ReReco 22Sep2023

The $e\mu$ requirement removes $Z \rightarrow ee/\mu\mu$ by construction, so the irreducible background is **real $t\bar{t}$** rather than Drell-Yan.

Reference: **AN-23-102**, the full Run 2 analysis at 138 fb⁻¹.

Signal and final state

$H \rightarrow WW \rightarrow 2\ell 2\nu$, produced with a charm quark. Final state: **opposite-sign** $e\mu$ + MET + ≥ 1 c-tagged jet

object	selection
muons	tight ID + tight PF iso, $p_T > 10$, $ \eta < 2.4$
electrons	wp80iso, $p_T > 10$, $ \eta < 2.5$, $\Delta R(e, \mu) > 0.4$
$\ell\ell$ pair	OS, $p_T > 20/10$, pair $p_T > 30$, $m_{\ell\ell} > 12$
jets	$p_T > 30$, $ \eta < 2.4$, tight-lepveto ID, $\Delta R(j, \ell) > 0.4$
c-jets	$p_T > 20$, PNet medium CvL/CvB
MET	PuppiMET

$e\mu$ removes the $Z \rightarrow ee/\mu\mu$ peak **by construction** — the irreducible background is **real $t\bar{t}$** , not Drell-Yan. Triggers: MuonEG, SingleMu, SingleEle.

Signal region composition

process	SR yield	share of SR
$t\bar{t}$	17,744	82.3%
st	1,543	7.2%
vjets	1,523	7.1%
diboson	609	2.8%
higgsbkg	132	0.6%
H+c signal	0.26	0.001%
total	21,552	

2 · MVA-defined regions



Six classes, one network

setting	value
model	<code>SimpleMLP_MultiClass</code> (b-hive)
classes	<code>hplusc, higgsbkg, tt, st, diboson, vjets</code>
features	26 , including the 11 c-tag one-hot categories
training	30 epochs, batch 1024, lr 1e-3, loss weighting on
split	80/20, deterministic by NanoAOD <code>event</code> id

The 11 one-hot categories mean the network sees the **full 2D CvL/CvB plane** — the same information the scale factors are binned in, not a binary pass/fail.

argmax defines the regions

The winning class picks the channel; the winning score is the discriminant.

region	events	dominant process
SR_hplusc	21,552	tt 82.3%
CR_higgsbkg	9,220	tt 86.4%
CR_tt	44,199	tt 94.1%
CR_st	20,135	tt 87.6%
CR_diboson	6,585	tt 72.4%
CR_vjets	9,214	vjets 43.2%

Regions are **orthogonal by construction** — no cut optimisation to tune or defend, and every event is used somewhere.

Four of five CRs are tt-dominated — by design

The SR is itself **82% tt**: in an $e\mu$ final state $t\bar{t}$ is dominant everywhere, so a tt-rich CR is expected, not a failure.

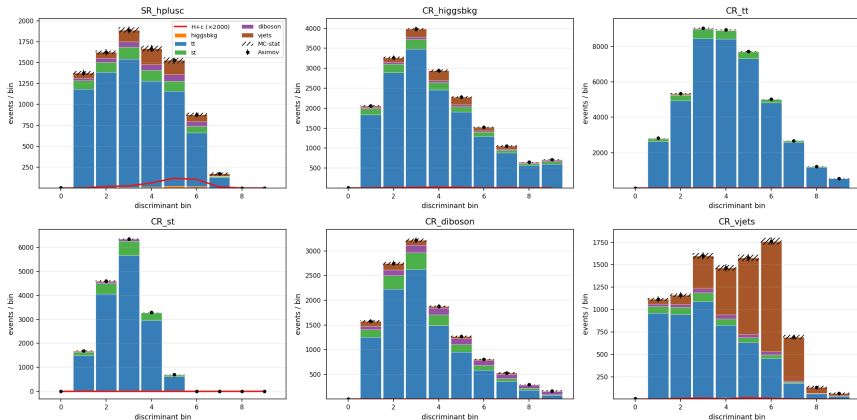
CR_tt: 94.1% purity over 44k events. It pins the free-floating `rate_tt`, which covers 82% of the SR — the single most valuable constraint in the fit.

CR_vjets holds 59.6% of all V+jets in the fit. Without it there is no V+jets constraint from anywhere.

Collapsing the CRs to single bins was **measured**: +33 units for three CRs, $\sim +50$ for all five. The shapes carry real constraint, so all six channels keep 10 bins.

Templates entering the fit

All combine input channels — stacked nominal templates (v11, 2022postEE)



6 channels $\times$ 6 processes $\times$ 10 bins. **CR_vjets** (bottom right) is the only region where V+jets (orange) is visible — it carries 59.6% of all V+jets in the fit.

3 · c-tagging

2D working points and scale factors

From working points to a plane

PNet gives **two** discriminants per jet: **CvL** (charm vs light) and **CvB** (charm vs bottom). A single working-point cut throws most of that away. The calibration instead partitions the **whole plane** into **11 categories** L0, C0–C4, B0–B4 — including the untagged L0 bin.

property	value
axes	CvL, CvB — verified sufficient , no third variable needed
categories	11: L0 , C0 – C4 , B0 – B4
index	computed from CvL/CvB already stored per jet
applied	natively in the processor (CTag2DCorrector)

Because the scheme spans the **full plane including** L0 , the SF machinery works even for a selection with no working point at all.

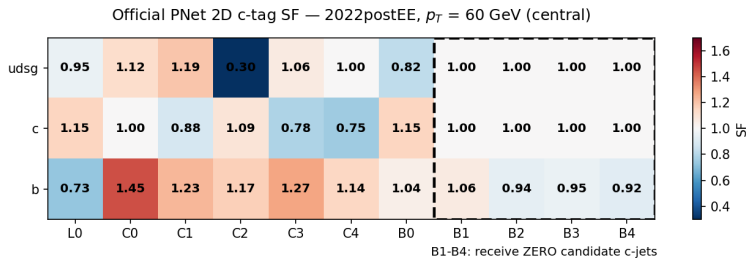
Only 7 of 11 categories are populated

category	jets	c (%)	b (%)	light (%)
L0	130,694	13.5	4.7	81.8
C0	188,702	28.2	6.2	65.7
C1	170,043	58.0	7.9	34.1
C2	17,823	58.6	33.4	8.0
C3	1,082	30.6	57.9	11.5

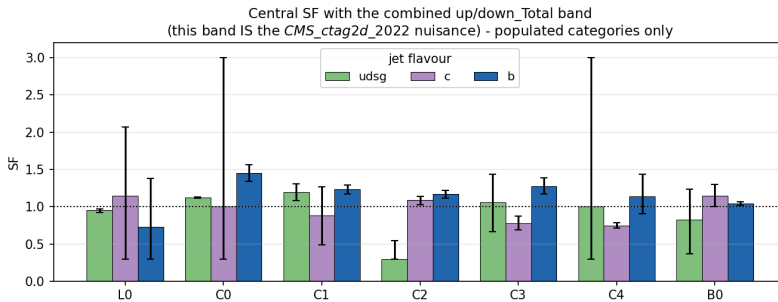
The scheme targets a broader phase space than ours — the b-rich B1 – B4 categories are empty after the $e\mu + c$ -jet selection.

Category index is computed from CvL/CvB already stored per jet — **no NanoAOD reprocessing was ever needed.**

The scale factor matrix



The uncertainty band is the nuisance



What the calibration costs

configuration	limit
no SF	1150
+ CMS_ctag2d_2022	1164

Applying the calibration **worsens** the limit by 14 units — as it must. A scale factor adds a nuisance. The point is correctness.

Currently **one nuisance** covering the whole plane. Decorrelation is pending.

4 · Negative-weight reweighting



full detail: `negrw-deck.pdf` (57 slides)

The estimator is a cancellation

aMC@NLO events carry **signed** weights. The yield estimator

$$\hat{N}_B = \sum_{i \in B} w_i, \quad w_i = \pm |w_i|$$

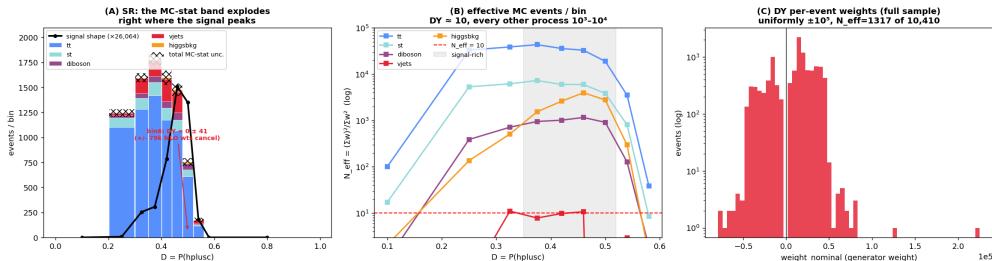
is a **difference of two large numbers**.

- ▶ the **expectation** is correct — the yield is unbiased
- ▶ the **variance** is not: it scales with $\sum w_i^2 = \sum |w_i|^2$, the *unsigned* statistics, while the mean is the much smaller signed sum
- ▶ effective statistics collapse: $N_{eff} = (\sum w)^2 / \sum w^2 \ll N$

16.4% of our V+jets events carried a negative weight. In a sparse bin the positives and negatives can cancel **completely** — leaving zero content with a large finite error.

V+jets was starved exactly under the signal

autoMCStats blow-up = DY (vjets) is starved in the H+c signal region



(A) the MC-stat band explodes where the signal peaks — one bin was $DY = 0 \pm 41$ from $\pm 79k$ weights cancelling · **(B)** N_{eff} per bin: V+jets ≈ 10 , every other process 10^3 - 10^4 · **(C)** DY per-event weights, uniformly $\pm 10^5$

The method — an algebraic identity

Write the signed generator density as a difference of two positive densities:

$$\text{PDF}(\vec{x}) = a \text{PDF}_+ - b \text{PDF}_-, \quad a - b = 1$$

With $P_+(\vec{x}) = a\text{PDF}_+ / (a\text{PDF}_+ + b\text{PDF}_-)$:

$$\text{PDF}(\vec{x}) = \underbrace{(2P_+(\vec{x}) - 1)}_{g(\vec{x})} \cdot (a\text{PDF}_+ + b\text{PDF}_-)$$

The right factor is the **unsigned** density — what $\sum |w|$ samples.

So $\sum |w| g$ and $\sum w$ estimate **the same distribution**. Yield-preserving by construction, not an approximation.

Exact for the *true* P_+ . We use $\hat{P}_+$ from a finite classifier — so closure becomes an **empirical** property that must be measured.

Training region — train loose, infer tight

$g(\vec{x})$ is a **generator property**; it does not know about the analysis selection. So we train on a far larger sample than the SR.

region	
train	all base cuts + veto of the $e\mu$ SR topology
infer	the tight $e\mu$ signal region

Disjoint by construction. An event that is not “exactly one $e\mu$ pair” can never be an SR event — no event-id bookkeeping needed.

Rejected alternatives: inverting MET (kinematically softer $\rightarrow$ extrapolation) and inverting the c-jet requirement (flips into the *opposite* gen corner).

Inputs — 20 generator-level features

group	variables
LHE event	<code>lhe_njets</code> , <code>lhe_nb</code> , <code>lhe_nc</code> , <code>lhe_nuds</code> , <code>lhe_nglu</code> , <code>lhe_npnlo</code>
LHE kinematics	<code>lhe_ht</code> , <code>lhe_htincoming</code> , <code>lhe_vpt</code> , <code>lhe_alphas</code>
gen partons	multiplicity, <code>n_pt20/100/200</code> , incoming PDG IDs
leading partons	<code>genparton1/2_pt</code> , <code>genparton1/2_eta</code>

All generator-level. The paper explicitly **rejects reco-level variables** — closure is only guaranteed on quantities the generator itself sampled.

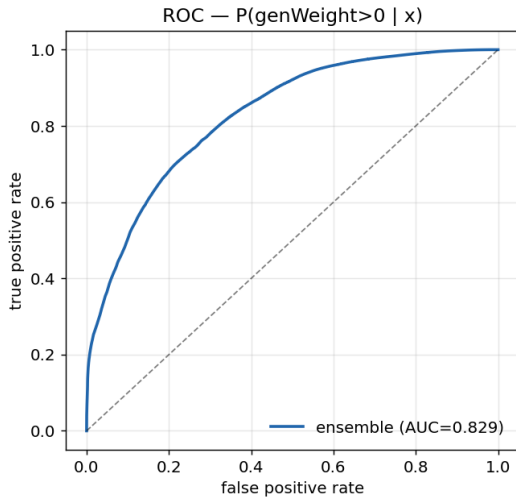
The two weight classes separate visibly in nearly every one of these features.

The classifier

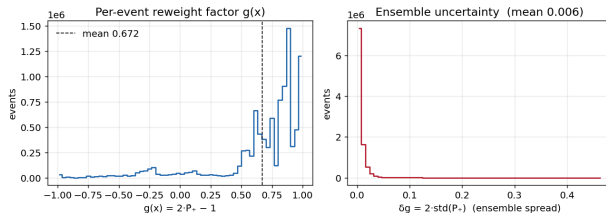
setting	value
model	<code>HistGradientBoostingClassifier</code> (scikit-learn)
inputs	20 generator-level (<code>lhe_*</code> , <code>genparton_*</code>)
training events	9.8M
ensemble AUC	0.829

AUC 0.83 on an **intrinsically stochastic** target — the weight sign is not a deterministic function of the kinematics, so a perfect classifier cannot exist even in principle. 0.83 means the generator-level features genuinely carry the information the NLO subtraction encodes.

Classifier ROC

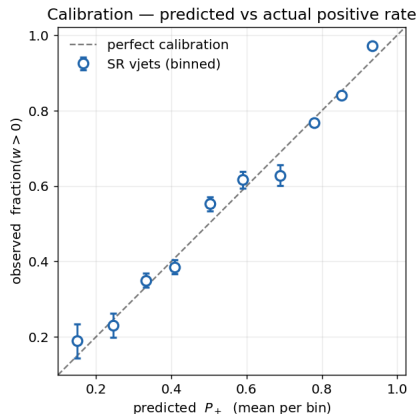


The reweight factor and its spread



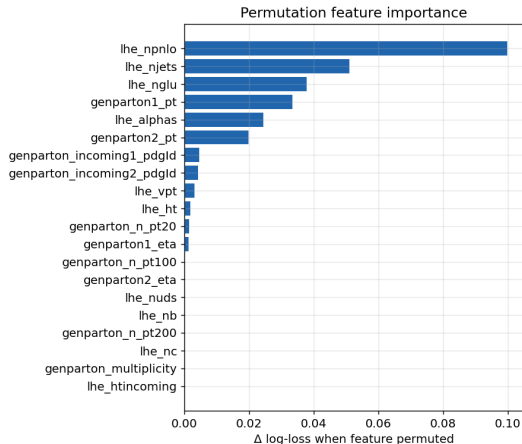
g mean **0.672** $= 2 \times 0.836 - 1$, matching the global positive fraction $\cdot$ range $[-0.991, 0.993]$, inside the physical $[-1, 1]$ $\cdot$ δg mean **0.006**, which becomes the `CMS_ngrw_vjets` nuisance.

Is P_+ calibrated outside the training region?



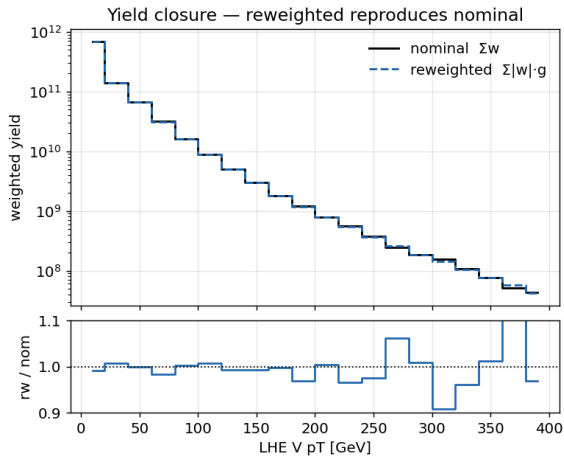
Bin SR events by **predicted** P_+ , plot the **observed** fraction with $w > 0$. Points land on the diagonal across the full range ($P_+ \approx 0.15 \rightarrow 0.93$) — the classifier is calibrated where it is applied, not only where it was trained.

Which features carry the information



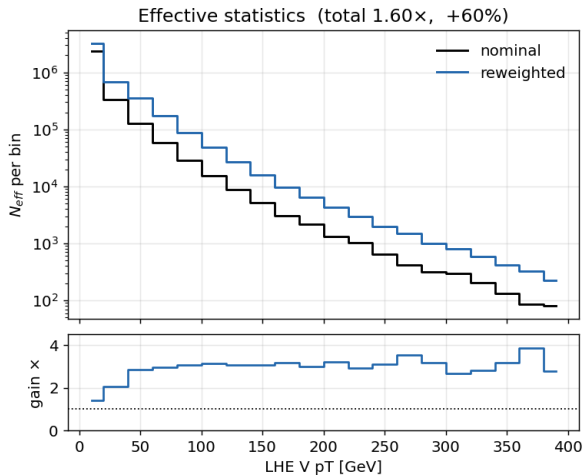
Permutation importance: the increase in log-loss when a feature is scrambled.

Closure



Training-region closure **0.994** on 9.8M events — the reweighting removes variance and nothing else.

The gain



Integrated gain $1.6\times$; per-bin ratio $\approx 3\times$ in the signal-rich bins.

SR closure and the renormalisation

process	$\sum w$	$\sum w g$	renorm
DY (both mass bins)	1.014e8	1.027e8	0.987
$W \rightarrow \ell \nu$	5.238e7	5.876e7	0.891

Applied **per dataset** in the card builder: the reweighted template is scaled back to the nominal yield, so the **yield is restored exactly** while the variance reduction — from per-bin spread, not the integral — is untouched.

These factors predate the W+jets sample change — they are for the old inclusive `WtoLNu_2Jets`. **The classifier will be retrained on the jet-binned 0J/1J/2J samples** and the factors re-derived.

5 · W +jets jet-binned samples



Inclusive → jet-binned

sample	xsec (pb)	events	files	neg-w
WtoLNu-2Jets_0J	55,760	678M	3,432	10.2%
WtoLNu-2Jets_1J	9,529	523M	2,669	25.7%
WtoLNu-2Jets_2J	3,532	345M	2,135	34.7%
sum	68,821	1,546M	8,236	
<i>old inclusive</i>	<i>67,710</i>	<i>282M</i>	<i>381</i>	<i>16.1%</i>

Cross sections from **XSDB**; sum within **+1.6%** of the inclusive — normal NLO merging spread, not a double-count. Measured negative-weight fractions match XSDB to $\sim 1\%$.

Why the gain is not simply $5.5\times$

Raw events rise $5.5\times$, but the useful quantity is effective statistics:

sample	n_{eff}/N	equivalent lumi
0J	0.635	7.7 /fb
1J	0.237	13.0 /fb
2J	0.094	9.1 /fb
combined		29.8 /fb
<i>inclusive</i>	<i>0.460</i>	<i>1.9 /fb</i>

The inclusive sample spends most of its cross section on **0-jet** events, which almost never survive the ≥ 1 c-jet requirement.

Of the 4,851 surviving SR events, **2J alone contributes 69%** — despite being the smallest sample. The jet-binned samples are enriched in exactly what the selection needs.

W+jets result: 1160 $\rightarrow$ 1034

	before	after
expected UL	1160	1034
V+jets n_{eff} (SR)	280	1170
V+jets stat. error	5.98%	2.92%
V+jets rate	1508	1523

The rate barely moved while the statistical error halved — the signature of a statistics gain, not a normalisation change.

6 · The combine card



Card structure

```
imax 6      # 6 channels: SR_hplusc + 5 argmax CRs
jmax 5      # 6 processes minus one
kmax *      # nuisances counted automatically

shapes * SR_hplusc v11_hplusc_2dcat.root \
    SR_hplusc_$PROCESS SR_hplusc_$PROCESS_$SYSTEMATIC
... one line per channel
```

channels	SR_hplusc , CR_higgsbkg , CR_tt , CR_st , CR_diboson , CR_vjets
processes	hplusc , higgsbkg , tt , st , diboson , vjets
binning	10 bins per channel — the winning MVA score
histograms	1,626 in the ROOT file (nominal + all up/down)
observation	-1 in every channel — Asimov, blind

The two special entries

A free-floating tt normalisation:

```
rate_tt  rateParam *  tt  1.0  [0,5]
```

One parameter, **shared across all six channels**, so CR_tt (94% pure, 44k events) determines the tt normalisation in the SR. This is why $t\bar{t}$ carries no `xsec` lnN — its rate comes from data, following AN-23-102. **Bin-by-bin MC statistics:**

```
SR_hplusc      autoMCStats 10
CR_higgsbkg    autoMCStats 10
... one line per channel
```

Barlow–Beeston-lite. Threshold 10 means bins with $n_{eff} > 10$ get a single Gaussian nuisance; sparser bins get individual Poisson terms.

How a template becomes a card entry

```
parquet -> read_scale = lumi x xsec / sumw -> TH1 per (channel, process)
                                                -> + one TH1 per systematic up/down
```

Normalisation is a two-place mechanism. The per-event `genWeight` enters post-selection through the weights container; the denominator `sumw` is summed **pre-selection** and written to `sumw_records` — including chunks that select zero events. Reading it from parquet metadata instead undercounts.

Object shifts (JES/JER, lepton scale/res) cannot be weights — they change the MVA score, so each is a **separate parquet tree**, re-selected and re-scored, giving 12 shifted directories.

6 · Systematics

The full inventory

22 shape · 9 lnN · rate_tt free rateParam · autoMCStats (threshold 10)

shape (weight)	shape (object shift)
pileup	CMS_scale_j_2022 (JES)
ps_isr, ps_fsr	CMS_res_j_2022 (JER)
scalevar_muR, _muF, _muR_muF	CMS_scale_e_2022, CMS_res_e_2022
lhe_pdf, lhe_alphaS	CMS_scale_m_2022, CMS_res_m_2022
muon_id, muon_iso	
electron_id, electron_reco ×3	lnN
CMS_ctag2d_2022	lumi_13p6TeV, xsec_st/diboson/vjets/higgsbkg
CMS_negrw_vjets	BR_HtoWW, BR_Htautau, xsec_hplusc_4FS_5FS
flavor_composition_ggH (placeholder)	

Object shifts need full reprocessing

JES/JER and lepton scale/resolution **change the MVA score**, so they cannot be applied as an event weight.

Each is a **separate parquet tree**, re-run through the full selection *and* re-scored by the network — 12 shifted directories.

Inference coverage over every shift directory is **verified after each rebuild** — a partially-scored set would leave object-shift templates frozen at nominal.

Top- p_T reweighting

property	value
form	theory-based NNLO/NLO
Run 3 factor	$\times (0.991 + 7.5\text{e-}5 \cdot p_T)$
nominal	reweighted
p_T cap	none — well-behaved at high p_T

```
sf_run2 = 0.103*exp(-0.0118*pT) - 0.000134*pT + 0.973
sf       = (0.991 + 0.000075*pT) * sf_run2
weight   = sqrt(prod(sf))           # over the two gen tops
down     = 1.0                     # "no correction" is the variation
```

Higgs heavy-flavour

Implemented as a **per-event weight instead**, ported from HiggsDNA `Higgs_plus_HF_syst` :

```
num_HF_jets = ak.sum(genJets.hadronFlavour == 4, axis=-1) # pT>25, |eta|<2.5
up    = where(num_HF_jets > 0, 1.5, 1.0)    # AN-23-102: 50% on ggH
down  = where(num_HF_jets > 0, 0.5, 1.0)
```

Keyed on **gen-jet flavour**, so process grouping stops mattering.

MET unclustered energy

Implemented, not yet in the card — it needs the shifted trees from a reprocessing pass, so it does not appear in the 22 shape nuisances above.

For Run 3 PuppMET the shift is taken directly from the NanoAOD branches:

```
events.PuppMET.ptUnclusteredUp / ptUnclusteredDown  
events.PuppMET.phiUnclusteredUp / phiUnclusteredDown
```

Matches HiggsDNA `MET_syst_Unclustered` — same branches, same up/down ordering.

What is implemented

systematic	implementation
CMS_negrw_vjets	ensemble spread of the negative-weight reweighting
CMS_ctag2d_2022	2D CvL/CvB SF, applied natively in the processor
electron_reco $\times 3$	p_T -binned (<20 , $20-75$, >75 GeV)
lhe_pdf , lhe_alphaS	NNPDF replicas, MC2Hessian
top_pt	hh2bbww theory form — <i>pending activation</i>
higgs_plus_c	per-event HF weight — <i>pending activation</i>
MET unclustered	object shift — <i>pending activation</i>

The last three are implemented and verified but **not in the current card** — each needs its weight column or shifted tree from a reprocessing pass.

Deliberately excluded

item	decision
muon reco SF	not applicable — HiggsDNA exposes no reco key; hh2bbww registers <code>mu_id_sf</code> / <code>mu_iso_sf</code> and no <code>mu_reco_sf</code> . Two frameworks, same conclusion.
PU jet ID	absent from both frameworks for Run 3
UE / tune	requires dedicated tune samples — cannot be a weight
<code>hdamp</code> , <code>mtop</code>	sample-based tt modelling — under consideration , see next slide

tt modelling: `hdamp` and `mtop`

term	what it is
<code>hdamp</code>	The Powheg damping parameter controlling the matching between the NLO matrix element and the parton shower. It sets how much hard radiation comes from the ME rather than the shower, so varying it changes the jet multiplicity and jet p_T spectrum of $t\bar{t}$.
<code>mtop</code>	The top-quark mass used in generation. Varying it (typically ± 1 GeV) shifts the $t\bar{t}$ kinematics and acceptance .

Neither can be applied as an event weight — both require **dedicated alternative samples** generated with the varied parameter.

Undecided whether to add these. They are standard in Run 3 $t\bar{t}$ analyses, but absent from AN-23-102 Table 16, and $t\bar{t}$ normalisation here is already taken from data via the free `rate_tt`.

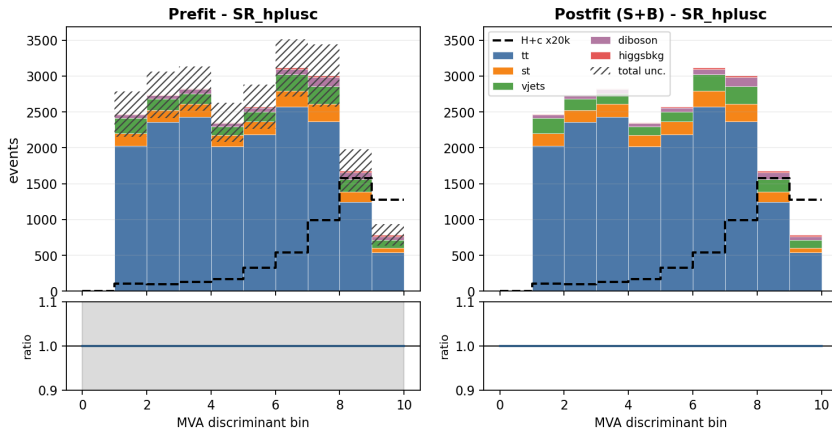
What moves the limit

frozen	limit	Δ	% of 1034
all constrained (stat-only)	641	393	38.0%
theory shapes (scale + PS + α_S)	883	151	14.6%
xsec_hplusc_4FS_5FS	921	113	10.9%
scalevar_muF	944	90	8.7%
autoMCStats (all)	956	78	7.5%
CMS_ctag2d_2022	964	70	6.8%
rate_tt	1011	23	2.2%
JES	1023	11	1.1%

autoMCStats is no longer the leading term — 7.5%, down from dominant before the W +jets change. Signal theory now leads.

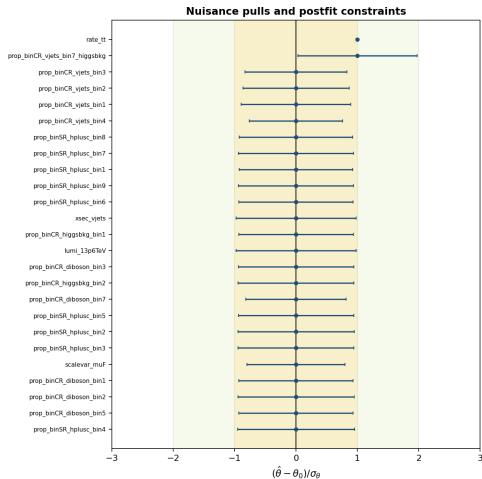
Rows overlap and do not add. `scalevar_muF` sits inside the theory group; freezing one nuisance also lets the others re-profile. Read this as a **ranking**, not a decomposition.

Template composition — signal region



Backgrounds stacked, $H+c$ overlaid ($\times 20k$). Hatched band = **prefit** total uncertainty, $\approx 14\%$ per bin. Postfit errors are absent because combine writes a **zero postfit covariance** on an Asimov fit.

Nuisance pulls and constraints



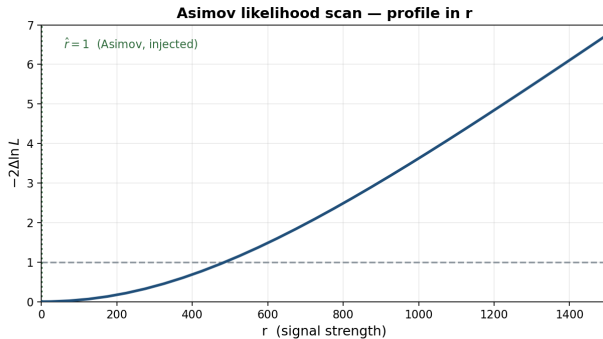
Reading the pulls

All **Gaussian** nuisances sit at **zero pull with $\sim$ unit width** — correct for an Asimov fit, and the check that nothing is pulled or over-constrained.

The two non-zero entries are **not Gaussian** and must not be read as pulls:

entry	value	why
rate_tt	1.000 ± 0.017	free rateParam, pinned by CR_tt
prop_binCR_vjets_bin7_higgsblog	1.000 ± 0.97	$n_{eff} = 2.1$, below the autoMCStats threshold $\rightarrow$ an individual Poisson term at its nominal scale

Likelihood scan



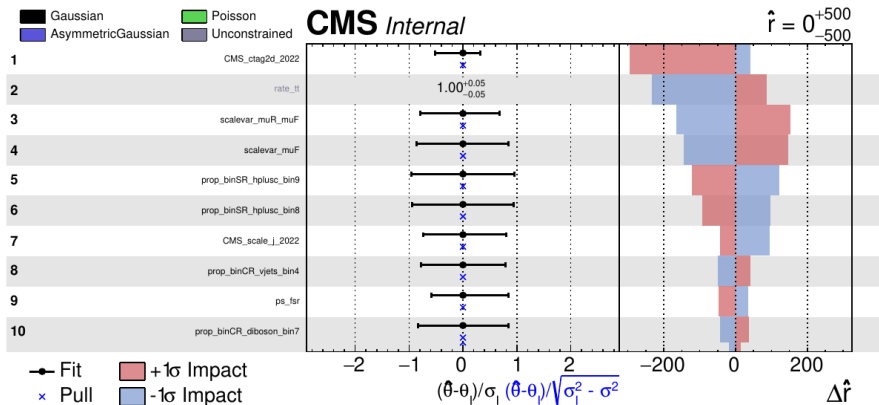
What the likelihood scan shows

The profile likelihood in the signal strength — all nuisances minimised at each point in r .

feature	value	meaning
minimum	$\hat{r} = 1$	the fit recovers the injected Asimov signal — no bias
$-2\Delta \ln L = 1$	$r \approx \mathbf{525}$	the 1σ interval on r
$-2\Delta \ln L = 3.84$	$r \approx \mathbf{1075}$	the 2σ crossing
curvature	—	how fast the likelihood falls away = the sensitivity

The 3.84 crossing (**1075**) sits close to the quoted CLs limit (**1034**) — the two constructions agree when the background-only and signal+background hypotheses are well separated, which is the regime here.

Nuisance impacts — top 10



The header $\hat{r} = 0^{+500}_{-500}$ is a `plotImpacts` rounding artefact — the actual fit returns $\hat{r} = 0.99996, -513/+483$, i.e. exactly the injected Asimov value.

Reading the impacts — the asymmetries

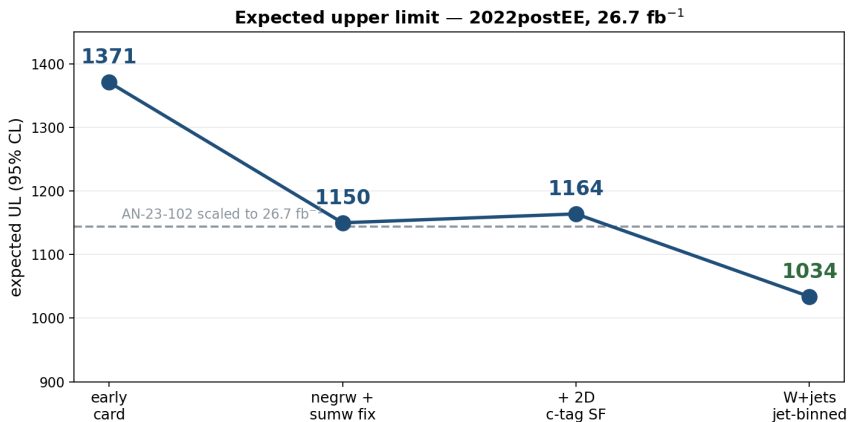
Red ($+1\sigma$) and blue (-1σ) are not mirror images. Three sources of asymmetry:

source	why
asymmetric lnN	<code>xsec_st</code> is 0.9873/1.0167 — the up and down variations differ by construction
template shape re-profiling	up/down shifts of JES or ctag move events between bins non-linearly pushing one nuisance lets the others re-fit, and they respond differently in each direction

The fit range matters: a symmetric range around $\hat{r}$ is required, or the -1σ impacts clip at the boundary and the plot comes out one-sided.

7 · Status and outlook

The road so far



Note the suppressed zero on the y-axis — the c-tag step is a genuine but small **+14** increase, not the spike it appears to be.

Where we stand

	limit
AN-23-102 scaled to 26.7 fb^{-1}	1144
this analysis	1034
our statistical only	641

Better than the published analysis scaled to our luminosity — 1034 vs 1144, a **9.6% improvement**, on **5× less data**.

The scaling is indicative. $503\sqrt{138/26.7} = 1144$ assumes a purely statistics-limited extrapolation; the AN is 73.8% statistical, so the true scaled value is **worse** than 1144 — our margin is **larger** than 9.6%.

What is still missing

item	status
negrw retrain	classifier to be retrained on the jet-binned W+jets samples
4FS/5FS signal theory	placeholder 1.30 — needs re-derivation at 13.6 TeV
Trigger scale factors	implemented, not yet enabled
Higgs heavy-flavour (<code>higgs_plus_c</code>)	implemented, pending activation
top-p_T reweighting	implemented, pending activation
Dataset substitutions	W+jets p_T -binned (full AN stitch) DY $\rightarrow$ Z $\rightarrow \tau\tau$ filtered tt / single-top <code>-ext</code> samples <code>WZto3LNu</code>

Priorities

#	item	why
1	Re-derive <code>xsec_hplusc_4FS_5FS</code>	30% flat lnN on a statistics-starved signal
2	Enable trigger SFs	implemented, one flag
3	Activate <code>higgs_plus_c</code> + <code>top_pt</code>	proper HF and tt treatment
4	Add the missing datasets	more MC statistics
5	Decorrelate <code>CMS_ctag2d_2022</code>	currently one nuisance over the whole plane

Summary

Expected UL 1371 $\rightarrow$ 1034 · statistical-only **641**

V+jets n_{eff} **280 $\rightarrow$ 1170**

Within $\sim 5\%$ of the Run 2 analysis scaled to our luminosity

Remaining: 4FS/5FS re-derivation, trigger SFs, dataset substitutions